

SI Session 2 Review

Owen Stayton

January 2026

1 Problems

1.1 Separable Variables

1.1.1 Problem 1

$$\frac{dy}{dx} = \frac{x \cos(x^2)}{\sin(y)}, \quad y(0) = \frac{\pi}{2}$$

1.1.2 Problem 2

$$\frac{dy}{dx} = y^2 \frac{1 - x^2}{x(1 + x^2)}$$

1.1.3 Problem 3

$$\frac{dy}{dx} = \frac{y + 1}{y} 2xe^{x^2}$$

1.2 Homogeneous

1.2.1 Problem 4

$$\frac{dy}{dx} = \frac{x + y}{x - y}$$

1.2.2 Problem 5

$$\frac{dy}{dx} = \frac{x + 2y}{2x - y}$$

1.2.3 Problem 6

$$\frac{dy}{dx} = \frac{x^2 + 3xy}{x^2 - xy}$$

1.3 Exact

1.3.1 Problem 7

$$(ye^{xy} + 2x)dx + (xe^{xy} + 3y^2)dy = 0$$

1.3.2 Problem 8

$$(2xy + y^2 + \sin(x))dx + (x^2 + 2xy - \cos(y))dy = 0$$

1.3.3 Problem 9

$$(3x^2y + \sin(y))dx + (x^3 + x \cos(y))dy = 0$$

1.4 Linear

1.4.1 Problem 10

$$\frac{dy}{dx} - \tan(x)y = \sec(x)$$

1.4.2 Problem 11

$$\frac{dy}{dx} + \frac{x}{x^2 + 1}y = \frac{x}{x^2 + 1}$$

1.4.3 Problem 12

$$\frac{dy}{dx} + \sec(x)y = \tan(x)$$